

1. a Neural Network for Multiclass Classification:

- A. a neural network is able to classify which multiple possible classes is a pattern belonged to
- B we can create one output perception per class and teach the neural network with multiple output perceptions to match those output values with those ground truths edited by one-hot encoding in the corresponding pair training example.

2. One-hot Encoding vs. Binary Encoding:

- A. one-hot encoding: a type of sparse encoding, implies only one number in a string of code is hot (singal is 1), and surplus numbers in a string of code are not hot (singal is 0) at the same time
- B. binary encoding: a type of dense encoding, means that it can reduce number of number of output perception in a neural network by binary machanism, but we have to possess a good mix of 1s and 0s, so that it is not the most suitable encoding for a neural network to do multiclass classification.
- C. one-hot encoding:

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1
Red	1	0	0

D. binary encoding:

Binary Mapping Scheme

Since there are 3 distinct classes (Red, Blue, Green), we need **2 bits**:

- Red → 00
- Blue → 01
- Green → 10

Binary Code Representation Table

Color	Binary Code (b_1b_0)
Red	00
Blue	01
Green	10